Integer quotients of ternary Cantor numbers: a power lower bound and an infinite exceptional family

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Abstract

Let $\mathcal{D}$ be the nonnegative integers whose ternary digits belong to $\{0,1\}$. We study positive integer quotients of members of $\mathcal{D}$. First, the number $M_{\mathrm{C}}(X)$ of integers $1 \le M \le X$ with $N, MN \in \mathcal{D}$ for some positive integer N satisfies

$$M_{\mathrm{C}}(X) \gg X^{\beta_*}, \qquad \beta_* = \frac{(1+s)^2}{7-5s} = 0.691726633148796\ldots, \quad s = \log_3 2.$$

Version 2 improves the original exponent $0.6660585297\ldots$, without an epsilon loss, by an endpoint mean-discrepancy estimate and an exact fixed-residue-class energy certificate. The same bound holds for the number of positive-dimensional intersections of the 3-adic Cantor set with its inverse integer dilates. Second, for every $k \ge 0$, the integer $621 \cdot 3^{4k} - 20$ is not such a quotient, although it avoids the standard lowest-digit and leading-digit obstructions. This proves that the sporadic exception set of Bai, Meleshko, Riasat and Shallit is infinite. The first proof combines Fourier sampling over Cantor-restricted denominators with a finite-word Fibonacci estimate and a sum over 3-adic distances. The second constructs an explicit predecessor-closed carry certificate of $38k + 50$ states for $k \ge 1$, with a separate certificate for $k = 0$. The original proofs and the version-2 improvement are retained in full. Version 3 proves that every member of this nonquotient family nevertheless has a positive-dimensional 3-adic Cantor intersection, with an explicit lower bound given by two return cycles of unequal lengths. It also proves that every finite representation of 289 has the same polynomial quotient $1 - t^2 - t^3 + t^4 + t^5$, answering the local-representation question in the specified version of Anders–Dawsey–Reznick–Sisneros-Thiry. An explicit family of local representations completes the new section.

1 The quotient problem and the main results

Version history. Version 1 (6 November 2026) established Theorems 1.1 and 1.2. Version 2 (8 November 2026) retains their full proofs and adds Theorem 9.1 in Section 9, raising the exponent from $0.6660585297\ldots$ to $0.691726633148796\ldots$ without an epsilon loss. The article title, sole author, and citation key `iuzhakov2026cantorquotients` are unchanged. Version 3 (12 November 2026) adds the positive-dimension theorem for the same sporadic family in Section 10, and the universal-numerator theorem for 289, with an explicit local family, in Section 11. These additions do not change the counting exponent. Earlier releases retain their own manuscripts and dates. The sharp growth of M_{C} and Erdős Problem 406 remain open here.

Put

$$\mathcal{D} = \left\{ \sum_{j \ge 0} \epsilon_j 3^j : \epsilon_j \in \{0,1\},\ \epsilon_j = 0 \text{ for all large } j \right\}, \qquad \mathcal{D}^+ = \mathcal{D} \setminus \{0\},$$

and let

$$\Sigma = \left\{ \sum_{j \geq 0} \epsilon_j 3^j : \epsilon_j \in \{0, 1\} \right\} \subset \mathbb{Z}_3.$$

Here $\mathbb{N}$ denotes the positive integers. Define the set of positive integer quotients and its counting function by

$$\mathcal{V} = \mathbb{N} \cap (\mathcal{D}^+ / \mathcal{D}^+), \qquad M_{\mathrm{C}}(X) = \#(\mathcal{V} \cap [1, X]) \quad (X \geq 1).$$

Equivalently, $M \in \mathcal{V}$ if some positive integer N has $N, MN \in \mathcal{D}$. Positivity of the witness is essential: allowing $N = 0$ would make this condition vacuous. For a positive integer M, also put

$$C(1, M) = \{x \in \mathbb{Z}_3 : x, Mx \in \Sigma\}, \qquad M_{\mathrm{H}}(X) = \#\{1 \leq M \leq X : \dim_H C(1, M) > 0\}.$$

1.1 A power lower bound

Lagarias asks for upper and lower bounds for these two counting functions in Problems 1.10 and 1.11 of [1], numbered Problems 1 and 2 in the arXiv version. Taking $N = 1$ gives the immediate lower bound of order $X^{\log_3 2}$. A positive integer witness also implies positive Hausdorff dimension by the nonoverlapping-block construction in [1]; we include this construction below. The inclusion of the integer-witness multipliers in the positive-dimensional multipliers is strict in general.

Theorem 1.1. *Let*

$$s = \log_3 2, \qquad \varphi = \frac{1 + \sqrt{5}}{2}, \qquad g = \log_3 \varphi, \qquad \alpha = \frac{s - g}{1 + g}, \qquad \beta = s + (2s - 1)\alpha.$$

There are absolute constants $c > 0$ and X_0 such that

$$M_{\mathrm{C}}(X) \geq cX^{\beta}, \qquad M_{\mathrm{H}}(X) \geq cX^{\beta} \quad (X \geq X_0).$$

In particular,

$$\alpha = 0.1341512347\ldots, \qquad \beta = 0.6660585297\ldots > \log_3 2.$$

The result improves the exponent in the immediate lower bound. It does not determine the sharp asymptotic size or positive density of either counting function. The distribution estimates used in its proof are proved directly, with their dependence on growing denominators explicit.

1.2 An infinite exceptional family

Following Bai, Meleshko, Riasat and Shallit [4, §1.4], define

$$\begin{aligned} \mathcal{O}_3 &= \{N \in \mathbb{N} : N \equiv 2 \cdot 3^{i-1} \pmod{3^i} \text{ for some } i \geq 1\}, \\ \mathcal{O}_\infty &= \mathbb{N} \cap \bigcup_{i \geq 0} [\tfrac{3}{2} 3^i,\, 2 \cdot 3^i], \\ \mathcal{F} &= \mathbb{N} \setminus (\mathcal{V} \cup \mathcal{O}_3 \cup \mathcal{O}_\infty). \end{aligned}$$

The first two sets encode the lowest nonzero digit and leading-digit obstructions to being a quotient. They are denoted $\mathcal{D}$ and $\mathcal{E}$ in [4]; our different symbols avoid conflict with $\mathcal{D}$, the digit-integer set. The original Cantor integers in that paper have digits $0, 2$. Their positive members form $2\mathcal{D}^+$, so their positive quotient set is exactly $\mathcal{D}^+/\mathcal{D}^+$. The sporadic set $\mathcal{F}$ therefore agrees with theirs.

Bai et al. ask whether $\mathcal{F}$ is infinite and propose the family in the following theorem.

Theorem 1.2. *For every integer $k \geq 0$,*

$$621 \cdot 3^{4k} - 20 \in \mathcal{F}.$$

In particular, there are no $a, b \in \mathcal{D}^+$ with $b = (621 \cdot 3^{4k} - 20)a$, and $\mathcal{F}$ is infinite.

The case $k = 4$ is verified in Jandhyala's versioned preprint [8, §6], which leaves the all-k assertion open. Its main theorems concern the different quotient problem for zero-free balanced digits $\{-1, 1\}$. Our proof of Theorem 1.2 uses the identity $b + 20a = 23 \cdot 3^{4k+3}a$ to obtain a binary delay register with a carry in a fixed set of 22 integers. An explicit predecessor-closed set separates the initial state from the only nontrivial predecessor of an accepting state.

1.3 Relation between the arguments

The quotient problem is also studied directly by Anders, Dawsey, Reznick and Sisneros-Thiry [5]. Multiplicative intersections and their automata are treated in [2, 3]. The analytic part of our counting proof is related to work on the distribution of integers with missing digits [6, 7], but no external uniformity estimate is used without proof.

The two original main theorems are logically independent. The first proves abundance among all positive integers; the second gives specified integers which fail to be quotients despite the elementary necessary conditions. The exceptional family is not a matching upper bound for $M_{\mathrm{C}}(X)$, and its nonrepresentation does not by itself determine $\dim_H C(1, M)$ for those multipliers. Section 10 supplies a separate explicit positive-dimension construction for every member of the family. Section 11 then studies universal and local finite representations through polynomial carry potentials.

Neither theorem concerns the number of exponents n for which $2^n \in \mathcal{D}$. In particular, neither proves the pointwise assertion in Erdős problem 406, excludes 1 from the infinite-visit exceptional set, or determines a finite-visit dimension limit. The arguments below use neither a general causal-container theorem nor a finite-rank unit-equation theorem. We make no claim of exhaustive priority verification.

2 Fourier sampling over reduced fractions

For $l \geq 0$, let

$$\mathcal{D}_l = \left\{ \sum_{j=0}^{l-1} \epsilon_j 3^j : \epsilon_j \in \{0, 1\} \right\}, \quad F_l(t) = 2^{-l} \sum_{z \in \mathcal{D}_l} \mathrm{e}(zt), \quad f_l(t) = |F_l(t)|,$$

where $\mathrm{e}(t) = \exp(2\pi i t)$ and $t \in \mathbb{T} = \mathbb{R}/\mathbb{Z}$. The product identity gives

$$F_l(t) = \prod_{j=0}^{l-1} \frac{1 + \mathrm{e}(3^j t)}{2}, \qquad f_{l+l'}(t) = f_l(t) f_{l'}(3^l t), \qquad 0 \leq f_l(t) \leq 1. \tag{1}$$

Lemma 2.1 (Elementary sampling)**.** *Let $0 < \delta \leq 1$ and let $T \subset \mathbb{T}$ be finite, with circular separation at least δ. If $P(t) = \sum_{j=0}^{K} c_j \mathrm{e}(jt)$, then*

$$\sum_{t \in T} |P(t)|^2 \leq (\delta^{-1} + 4\pi K) \sum_{j=0}^{K} |c_j|^2.$$

Proof. Center pairwise disjoint half-open arcs I_t of length δ at the points of T. For $G = |P|^2$ and $u \in I_t$,

$$G(t) \le G(u) + \int_{I_t} |G'(v)|\, dv.$$

Integrating in u and summing in t bounds the left side by $\delta^{-1}\|P\|_2^2 + \int_{\mathbb{T}} |G'|$. Since $\int |G'| \le 2\|P\|_2\|P'\|_2 \le 4\pi K\|P\|_2^2$, Parseval proves the assertion. □

Lemma 2.2. *For every $h \ge 0$,*

$$\int_{\mathbb{T}} |F_h|^2 = 2^{-h}, \qquad \int_{\mathbb{T}} |F_h|^4 = (3/8)^h.$$

The degrees of F_h and F_h^2 are $(3^h - 1)/2$ and $3^h - 1$.

Proof. The second moment follows from the distinct frequencies in $\mathcal{D}_h$. For the fourth moment, the coefficients of F_h^2 count sums of two elements of $\mathcal{D}_h$. At each ternary position the sums $0, 1, 2$ have multiplicities $1, 2, 1$. There is no carry, so the sum of the squares of the unnormalized coefficients is $(1^2 + 2^2 + 1^2)^h = 6^h$. The normalization is 16^h. The degree statements follow from $\max \mathcal{D}_h = (3^h - 1)/2$. □

For real $D \ge 2$, write

$$T_D = \{v/d \bmod 1 : 2 \le d \le D,\ (d, 3) = 1,\ 1 \le v < d,\ (v, d) = 1\}.$$

Distinct points of T_D have circular separation at least D^{-2}.

Lemma 2.3 (Farey Fourier bound)**.** *Put $\eta = \log_3(27/16) < s$. There is an absolute constant C such that, if $D \ge 2$ and $L \ge 3\lceil 2 \log_3 D \rceil$, then*

$$\sum_{t \in T_D} f_L(t) \le C D^{\eta}. \tag{2}$$

For each fixed D, the sum on the left tends to zero as $L \to \infty$.

Proof. Take $h = \lceil 2 \log_3 D \rceil$, so $D^2 \le 3^h < 3D^2$. Lemmas 2.1 and 2.2 give

$$\sum_{t \in T_D} f_h(t)^2 \le C(3/2)^h, \qquad \sum_{t \in T_D} f_h(t)^4 \le C(9/8)^h.$$

Cauchy–Schwarz yields

$$\sum_{t \in T_D} f_h(t)^3 \le C(27/16)^{h/2} \le C' D^{\eta}. \tag{3}$$

Let $p = \lfloor L/h \rfloor \ge 3$. By (1), after discarding the remainder of length $L - ph$, Hölder's inequality gives

$$\begin{aligned} \sum_{t \in T_D} f_L(t) &\le \prod_{j=0}^{p-1} \left(\sum_{t \in T_D} f_h(3^{jh} t)^p \right)^{1/p} \\ &= \sum_{t \in T_D} f_h(t)^p \le \sum_{t \in T_D} f_h(t)^3. \end{aligned}$$

Multiplication by 3^{jh} permutes T_D, since its reduced denominators are coprime to 3. This proves (2). For a fixed nonzero $v/d \in T_D$, the orbit under multiplication by 3 is periodic and avoids zero. The product in (1) therefore tends to zero in modulus. Summing over the fixed finite set proves the last assertion. □

3 Distribution for most Cantor denominators

For $Y \geq 3$, set

$$S_Y = \{N \in \mathcal{D} : 1 \leq N \leq Y,\ N \equiv 1 \pmod 3\}.$$

Elementary digit counting gives

$$\tfrac{1}{4} Y^s \leq |S_Y| \leq 2Y^s. \tag{4}$$

For the lower bound use all words of length $\lfloor \log_3 Y \rfloor$ whose low digit is 1. For the upper bound use all words of length $K = \lfloor \log_3 Y \rfloor + 1$.

Lemma 3.1 (Divisor multiplicity)**.** *For integers $1 \leq d \leq Y$,*

$$T_Y(d) := \#\{N \in S_Y : d \mid N\} \leq 4(Y/d)^s.$$

Proof. Put $j = \lfloor \log_3 d \rfloor$. Fixing the digits at positions $j, \ldots, K-1$ covers $\mathcal{D}_K$ by 2^{K-j} intervals, each of length $(3^j - 1)/2 < d$. Each interval contains at most one multiple of d. The inequalities $3^K \leq 3Y$ and $3^j > d/3$ give $2^{K-j} < 4(Y/d)^s$. □

For $3 \nmid N$, define

$$R_N(L) = \sum_{v=1}^{N-1} f_L(v/N), \qquad R_1(L) = 0.$$

Proposition 3.2 (Average discrepancy on restricted denominators)**.** *As $Y, L \to \infty$ subject to $L \geq 3\lceil 2 \log_3 Y \rceil$,*

$$\sum_{N \in S_Y} R_N(L) = o(Y^s). \tag{5}$$

Consequently, for all sufficiently large such Y, L, at least $Y^s/8$ elements $N \in S_Y$ satisfy $R_N(L) \leq 1/2$.

Proof. Reducing the fractions v/N gives the exact identity

$$\sum_{N \in S_Y} R_N(L) = \sum_{\substack{2 \leq d \leq Y \\ (d,3)=1}} T_Y(d) \sum_{\substack{1 \leq v < d \\ (v,d)=1}} f_L(v/d).$$

Use dyadic bins $(2^{j-1}, 2^j]$ for $1 \leq j \leq J = \lfloor \log_2 Y \rfloor$, followed by the possibly empty bin $(2^J, Y]$. By Lemmas 3.1 and 2.3, the contribution of a complete bin, divided by Y^s, is bounded by

$$C 2^{-js} \sum_{t \in T_{2^j}} f_L(t) \leq C' 2^{-j(s-\eta)}.$$

The latter bounds are summable, since $\eta < s$. For each fixed j, the actual contribution tends to zero, by the last assertion of Lemma 2.3. First make the tail of the summable bounds small, and then take L large enough for the finitely many remaining bins. The final partial bin is at most $CY^{\eta - s}$ after division by Y^s, and also tends to zero. This proves (5). At most $2\sum_N R_N(L) = o(Y^s)$ denominators have $R_N(L) > 1/2$; combine this with (4). □

Proposition 3.3 (First moment)**.** *Let $L = \lfloor \log_3 X \rfloor$, and suppose $Y \to \infty$ with $L \geq 3\lceil 2 \log_3 Y \rceil$ and $X^s Y^{s-1} \to \infty$. There is a subset $G \subset S_Y$ such that*

$$\sum_{N \in G} |A_N(X)| \geq \tfrac{1}{64} X^s Y^{2s-1}, \qquad A_N(X) = \{1 \leq M \leq X : NM \in \mathcal{D}\},$$

for all sufficiently large X.

Proof. Choose $G = \{N \in S_Y : R_N(L) \le 1/2\}$, which has size at least $Y^s/8$. For each $N \in G$ put $L_N = \lfloor \log_3(NX) \rfloor \ge L$. By (1), $R_N(L_N) \le R_N(L) \le 1/2$. Orthogonality of additive characters modulo N gives

$$\#\{z \in \mathcal{D}_{L_N} : N \mid z\} \ge \frac{2^{L_N}}{N}(1 - R_N(L_N)) \ge \frac{2^{L_N-1}}{N}.$$

Every such z is at most NX. Removing zero and dividing by N yields

$$|A_N(X)| \ge \tfrac{1}{4} X^s N^{s-1} - 1.$$

Since $s < 1$, summing and using (4) gives

$$\sum_{N \in G} |A_N(X)| \ge \tfrac{1}{32} X^s Y^{2s-1} - 2Y^s \ge \tfrac{1}{64} X^s Y^{2s-1}.$$

□

4 A finite Fibonacci bound for overlaps

For $r \in 1 + 3\mathbb{Z}_3$, $r \ne 1$, put $e = v_3(r-1) \ge 1$. Define

$$W_n(r) = \{x \in \mathcal{D}_n : rx \bmod 3^n \in \mathcal{D}_n\},$$

using representatives in $[0, 3^n)$ for reduction. Let $H_n(e)$ be the number of independent sets of the graph on $\{0, \dots, n-1\}$ whose edges join positions at distance e.

Lemma 4.1 (Finite-word bound)**.** *For every $n \ge 0$,*

$$|W_n(r)| \le H_n(e) \le \varphi^{n+e}.$$

The bound applies to all locally admissible words, including those which have no infinite admissible continuation.

Proof. Write $r = 1 + 3^e u$, where $u \in \mathbb{Z}_3^\times$. Write the input digits as $x_j \in \{0, 1\}$ and the digits of ux as $y_j \in \{0, 1, 2\}$. In the addition $x + 3^e ux$, let $\gamma_q \in \{0, 1\}$ be the incoming carry at position q. For $q \ge e$, put $B_q = y_{q-e} + \gamma_q$. The allowed values and next carries are exactly

B_q	allowed x_q	γ_{q+1}
0	0, 1	0
1	0	0
2	1	1
3	0, 1	1

Thus $\gamma_{q+1} = \mathbf{1}_{B_q \ge 2}$ on admissible paths, independently of the chosen admissible value of x_q. For $q < e$ the carry is zero.

Fix $x_{<i}$. The preceding identity determines γ_{i+e} from $y_0, \dots, y_{i-1}$, hence from $x_{<i}$, independently of the intermediate admissible digits. Also $y_i \equiv u_0 x_i + R_i(x_{<i}) \pmod 3$, with $u_0 \in \{1, 2\}$. The output at position $i + e$ consequently has the form

$$x_{i+e} + u_0 x_i + Q_i(x_{<i}) \pmod 3. \tag{6}$$

At least one of $x_i = 0, 1$ makes the offset in (6) nonzero modulo 3, leaving at most one allowed value of x_{i+e}, regardless of the intermediate admissible digits.

Mark that future position as forced. A mark means that each subsequent admissible prefix on the branch has at most one child at the marked position; it does not specify a digit common to different branches. Maintain the set S of future marks. At a marked position there is at most one child. At an unmarked position with one child, retaining S suffices for an upper bound. At an unmarked position i with two children, one child retains S, while the other has the additional mark $i+e$, if that position is less than n. Additional restrictions may be discarded.

Backward induction now compares the number of completions with independent sets of the remaining distance-e graph, with positions in S required absent. At an unmarked vertex i, its deletion recursion is the sum of the counts without and with the additional mark $i+e$; at a marked vertex there is a single continuation. Requiring additional positions absent can only decrease the count. This proves $|W_n(r)| \le H_n(e)$, starting with no marks.

The graph is a union of at most e paths, one in each residue class modulo e. A path of m vertices has F_{m+2} independent sets, and $F_{m+2} \le \varphi^{m+1}$ by the Fibonacci recurrence. Multiplying these inequalities gives $H_n(e) \le \varphi^{n+e}$. □

Lemma 4.2 (Valuation summation)**.** *For $Y \ge 3$,*

$$\sum_{\substack{N<N' \\ N,N' \in S_Y}} \varphi^{v_3(N'-N)} \ll Y^{2s}.$$

Proof. Put $K = \lfloor \log_3 Y \rfloor + 1$. For fixed $N \in S_Y$ and $1 \le e < K$,

$$\#\{N' \in S_Y : N' \equiv N \pmod{3^e}\} \le 2^{K-e} \le 2Y^s 2^{-e}.$$

No distinct pair has valuation at least K. Bounding the number of pairs of valuation exactly e by the number with valuation at least e, and using $|S_Y| \le 2Y^s$, proves

$$\sum_{N<N'} \varphi^{v_3(N'-N)} \le 2Y^{2s} \sum_{e \ge 1} (\varphi/2)^e \ll Y^{2s}.$$

□

Proposition 4.3 (Overlap bound)**.** *For $X \ge 1$, $Y \ge 3$ and any $G \subset S_Y$,*

$$\sum_{\substack{N<N' \\ N,N' \in G}} |A_N(X) \cap A_{N'}(X)| \ll X^g Y^{2s+g}.$$

Proof. For a distinct pair, put $r = N'/N$ and $e = v_3(N'-N) = v_3(r-1) \ge 1$. If $M \in A_N \cap A_{N'}$, then $x = NM \le XY$ and $x, rx \in \mathcal{D}$. The assignment $M \mapsto x$ is injective. Taking $n = \lceil \log_3(XY+1) \rceil$ and applying Lemma 4.1 gives

$$|A_N \cap A_{N'}| \le \varphi^{n+e} \ll (XY)^g \varphi^e.$$

Sum this using Lemma 4.2. □

5 Proof of the power lower bound

Proof of Theorem 1.1. Take $Y = c_0 X^\alpha$, where $c_0 > 0$ is a sufficiently small fixed constant. We have $0 < \alpha < 1/6$. To check the latter without numerical rounding, note that $s < 7/11$ since $2^{11} < 3^7$, and $g > 3/7$ since $\varphi^7 = 13\varphi + 8 > 27$. The ratio $(s-g)/(1+g)$ is increasing in s and decreasing in g, so $\alpha < 8/55 < 1/6$. Consequently $L = \lfloor \log_3 X \rfloor \ge 3\lceil 2 \log_3 Y \rceil$ for large X. Also $X^s Y^{s-1} \to \infty$, since $s > 1/2$ and $\alpha < 1/6$.

Proposition 3.3 provides a set $G \subset S_Y$ with first moment at least $X^s Y^{2s-1}/64$. By Proposition 4.3, the sum of pairwise overlaps is at most $C_0 X^g Y^{2s+g}$. The ratio of this upper bound to the first-moment lower bound is

$$64 C_0 X^{g-s} Y^{1+g} = 64 C_0 c_0^{1+g}.$$

Choose c_0 so that this is at most $1/2$. Bonferroni's inequality gives

$$M_{\mathrm{C}}(X) \geq \left| \bigcup_{N \in G} A_N(X) \right| \geq \frac{1}{128} X^s Y^{2s-1} = \frac{c_0^{2s-1}}{128} X^{\beta}.$$

Finally, if $N, MN \in \mathcal{D}$ with $N > 0$, choose B with $3^B > \max(N, MN)$. For every sequence $(\epsilon_j) \in \{0,1\}^{\mathbb{N}_0}$,

$$x = N \sum_{j \geq 0} \epsilon_j 3^{Bj}$$

and Mx consist of nonoverlapping copies of the finite digit blocks for N and MN. Thus both belong to Σ. This embeds a scaled block Cantor set, of dimension $\log 2/(B \log 3) > 0$, into $C(1, M)$. Hence $M_{\mathrm{H}}(X) \geq M_{\mathrm{C}}(X)$, proving the second bound. □

6 A delayed digit equation

Fix $k \geq 0$ and put

$$m = 4k + 3, \qquad M = 23 \cdot 3^m - 20.$$

Suppose, for a contradiction, that $Ma = b$ with $a, b \in \mathcal{D}^+$. Write their finite ternary expansions as $a = \sum_{j \geq 0} a_j 3^j$ and $b = \sum_{j \geq 0} b_j 3^j$, extending both digit sequences by zeros. For negative indices set $a_j = 0$. The identity $b + 20a = 23 \cdot 3^m a$ gives the delayed digits $x_j = a_{j-m}$ and the recurrence

$$d_0 = 0, \qquad d_{j+1} = \frac{d_j + 20 a_j + b_j - 23 x_j}{3}. \tag{7}$$

Lemma 6.1. *Every d_j in (7) is an integer in $\mathcal{I} = \{-11, -10, \ldots, 10\}$, and $d_j = 0$ for all sufficiently large j.*

Proof. Set $f_h = 20 a_h + b_h - 23 a_{h-m}$. The finite identity $\sum_{h \geq 0} f_h 3^h = 0$ implies

$$d_j = 3^{-j} \sum_{h < j} f_h 3^h \in \mathbb{Z},$$

since the omitted terms are divisible by 3^j. This formula gives (7) and is zero once the partial sum includes every nonzero term. If $d_j \in \mathcal{I}$, its next numerator lies between -34 and 31. Since the numerator is divisible by 3, it lies between -33 and 30, so $d_{j+1} \in \mathcal{I}$. Induction starts at 0. □

Encode the last m input digits by the *binary* register

$$A_j = \sum_{i=0}^{m-1} a_{j-m+i} 2^i.$$

The oldest digit is in bit 0, so

$$x_j = A_j \bmod 2, \qquad A_{j+1} = \lfloor A_j/2 \rfloor + a_j 2^{m-1}. \tag{8}$$

This binary encoding is only a convenient label for a queue of digits. It is not the ordinary ternary value of that queue. The state space is

$$\{0, \dots, 2^m - 1\} \times \mathcal{I}.$$

The initial state is $(0, 0)$; a finite witness gives a finite path back to $(0, 0)$ containing an input digit 1.

Lemma 6.2. *Any positive witness gives a path from* $(0, 0)$ *to* $(0, -1)$*. Consequently, no positive witness exists if there is a set of states containing* $(0, -1)$*, closed under all predecessors, and omitting* $(0, 0)$*.*

Proof. Because $M \equiv 1 \pmod 3$, the same power of 3 divides a and b. Dividing it out preserves their allowed digits, so we may assume $a_0 = b_0 = 1$. The first state after $(0, 0)$ is then $(2^{m-1}, 7)$, which is different from $(0, 0)$.

Consider the first subsequent return to $(0, 0)$. A transition into this state has $a_j = 0$ and $A_j = x_j \in \{0, 1\}$ by (8). Equation (7) gives $d_j = 23x_j - b_j$. If $x_j = 1$, this value is 22 or 23, outside $\mathcal{I}$. If $x_j = 0$, the predecessor is $(0, 0)$ or $(0, -1)$. The first possibility is excluded at the first return. Thus the path passes through $(0, -1)$. A predecessor-closed set containing that state must contain every earlier state on the path, including $(0, 0)$, giving the asserted contradiction. □

For use in an explicit certificate, all predecessors can be written without a search. For a current state (A, d) put $\alpha = \lfloor A/2^{m-1} \rfloor \in \{0, 1\}$. For each $x, \beta \in \{0, 1\}$ the potential predecessor is

$$A^- = 2A - \alpha 2^m + x, \qquad d^- = 3d - 20\alpha - \beta + 23x. \tag{9}$$

It is a valid predecessor exactly when $d^- \in \mathcal{I}$. Here α is the input digit of the forward transition and β is its output digit. Formula (9) enumerates all choices; automatically $0 \le A^- < 2^m$.

7 The predecessor certificate for all $k \ge 1$

Assume now $k \ge 1$. Define

$$T_j = \frac{16^j - 1}{5} \quad (j \ge 1), \qquad V = \frac{3T_{k+1} - 1}{2}. \tag{10}$$

The numbers T_j are integers satisfying $T_{j+1} = 16T_j + 3$; their binary expansions are repetitions of the block 0011 after padding to length $4j$.

The following finite sets specify the allowed carries in the certificate:

$$\begin{aligned}
Z &= \{-10, -9, -4, -3, -1\}, & O &= \{-8, -7, -5, -4, 10\},\\
P &= \{-2, -1, 1, 2, 7, 8, 10\}, & P^+ &= P \cup \{4, 5\},\\
Q &= \{-7, -6, -4, -3, 2, 3, 5, 6\}, & R &= \{-10, -9, 5, 6, 8, 9\},\\
S &= \{1, 2, 4, 5, 10\}, & U &= \{-8, -7, -5, -4\},\\
U^+ &= U \cup \{-6, -3\}, & W &= \{2, 3, 5, 6\},\\
W^+ &= W \cup \{4, 7\}, & H &= \{5, 6, 8, 9\},\\
I_0 &= \{-10, -9\}, & J_0 &= \{-11, -6, -5, -3, -2\},\\
K_0 &= \{-10, -9, -7, -6, 8, 9\}, & N &= U.
\end{aligned}$$

The carry-set symbols in this display are local to the certificate. Distinct row names identify different registers, even when their carry sets agree.

Table 1: Definition of $\mathcal{S}_m$. Empty index ranges are omitted.

Row	Index range	Register A	Carry set
Z_0		0	Z
O_0		1	O
P_j	$1 \le j \le k$	T_j	P
P_{k+1}		T_{k+1}	P^+
Q_j	$1 \le j \le k+1$	$2T_j$	Q
R_j	$1 \le j \le k$	$4T_j$	R
S_j	$1 \le j \le k$	$4T_j + 1$	S
U_j	$1 \le j < k$	$8T_j + 1$	U
U_k		$8T_k + 1$	U^+
W_j	$1 \le j < k$	$8T_j + 2$	W
W_k		$8T_k + 2$	W^+
H_j	$1 \le j \le k$	$T_{j+1} + 1$	H
I_0^{row}		$T_{k+1} - 1$	I_0
I_1^{row}		$V - 1$	I_0
J^{row}		V	J_0
K^{row}		$V + 1$	K_0
N^{row}		$2T_{k+1} - 1$	N

Proposition 7.1. *The set $\mathcal{S}_m$ defined in Table 1 is closed under all predecessors in* (9)*. It contains* $(0, -1)$ *and omits* $(0, 0)$*. Its cardinality is* $38k + 50$*.*

Proof. We first determine the most significant bit of every register. Put $X = 16^k \ge 16$, so $2^{m-1} = 4X$ and $2^m = 8X$. For $j \le k$, $T_j < X/5$. All rows with most significant bit zero are bounded by

$$T_{k+1} + 1 = \frac{16X + 4}{5} < 4X; \qquad 8T_j + 2 \le \frac{8X + 2}{5} < 4X.$$

The rows $Q_{k+1}, I_1^{\mathrm{row}}, J^{\mathrm{row}}, K^{\mathrm{row}}, N^{\mathrm{row}}$ have respective registers

$$\frac{32X - 2}{5}, \quad \frac{24X - 9}{5}, \quad \frac{24X - 4}{5}, \quad \frac{24X + 1}{5}, \quad \frac{32X - 7}{5}.$$

Each lies in $[4X, 8X)$, so its most significant bit is one. This also verifies that every row defines a state in the prescribed space.

Table 2 gives the complete predecessor calculation. An unannotated destination means its whole carry set. A qualifier such as "base U" means the displayed smaller carry set, even when the destination row uses U^+. A dash denotes an empty image. Every entry allows both values $\beta = 0, 1$ and retains only $d^- \in \mathcal{I}$.

For the carry column, each entry is the finite calculation

$$\{3d - 20\alpha - \beta + 23x : d \text{ in the current carry set},\ \beta \in \{0, 1\}\} \cap \mathcal{I}.$$

The register column follows from $T_{j+1} = 16T_j + 3$ and the two boundary identities

$$4T_{k+1} - 2^m = V, \qquad 2V - 2^m = 8T_k. \tag{11}$$

For example, the row Q_{k+1} has register $2T_{k+1}$ and $\alpha = 1$. Its predecessor registers are $V, V + 1$, and its carry images are J_0, K_0, respectively. The row J^{row} has $\alpha = 1$; for $x = 1$ it maps to register $8T_k + 1$ and carries $\{-7, -6, -4, -3\} \subseteq U^+$. These boundary images explain the extra carries in P^+, U^+, W^+.

Thus every possible predecessor lies in a listed row. This verifies closure for all $k \ge 1$ using fixed finite carry sets and the exact identities (10) and (11). No upper bound on k

Table 2: All predecessor images from (9).

Current row	α	$x=0$	$x=1$
Z_0	0	$Z_0 : \{-10,-9,-4,-3\}$	O_0
O_0	0	—	P_1 (base P)
P_j, all j	0	Q_j	—
Q_j, $j \le k$	0	R_j	S_j
Q_{k+1}	1	J^{row}	K^{row}
R_j	0	—	U_j (base U)
S_j	0	W_j (base W)	—
U_j, $j < k$	0	—	P_{j+1} (base P)
U_k	0	I_0^{row}	P_{k+1} (P^+)
W_j, all j	0	H_j	—
H_j	0	—	—
I_0^{row}	0	—	N^{row}
I_1^{row}	1	—	—
J^{row}	1	—	$U_k : \{-7,-6,-4,-3\}$
K^{row}	1	$W_k : \{3,4,6,7\}$	—
N^{row}	1	—	I_1^{row}

is used. Since Z contains -1 and excludes 0, and no other row has register zero, the two required inclusions follow.

For the cardinality, the registers are distinct. In the interior,

$$T_j < 2T_j < 4T_j < 4T_j + 1 < 8T_j + 1 < 8T_j + 2 < T_{j+1},$$

and $T_{j+1}+1$ lies strictly between T_{j+1} and $2T_{j+1}$. At the final boundary, writing $T = T_{k+1} \ge 51$, the additional order is

$$T - 1 < T < T + 1 < V - 1 < V < V + 1 < 2T - 1 < 2T.$$

The register 1 lies below $T_1 = 3$. Summing the sizes of the carry sets in Table 1 now gives $38k + 50$. □

Lemma 6.2 and Proposition 7.1 exclude a positive witness for every $k \ge 1$.

8 The base case and the sporadic exceptional set

For $k = 0$ we have $m = 3$ and $M = 601$. The small certificate in Table 3 replaces Table 1.

Table 3: A predecessor-closed certificate of 65 states for $m = 3$.

Register A	Carry values
0	$\{-10,-9,-4,-3,-1,3,4,6,7,8,9\}$
1	$\{-8,-7,-6,-5,-4,-3,10\}$
2	$\{-10,-9,3,4,6,7\}$
3	$\{-10,-9,-2,-1,1,2,4,5,7,8,10\}$
4	$\{-11,-6,-5,-3,-2,8,9\}$
5	$\{-10,-9,-8,-7,-6,-5,-4,8,9\}$
6	$\{-7,-6,-4,-3,2,3,5,6\}$
7	$\{-10,-9,-8,-7,-5,-4\}$

Indeed, for each entry (A, d) and $x, \beta \in \{0, 1\}$, put

$$\alpha = \lfloor A/4 \rfloor, \qquad A^- = 2A - 8\alpha + x, \qquad d^- = 3d - 20\alpha - \beta + 23x.$$

Whenever $-11 \le d^- \le 10$, the value d^- occurs in the row A^- of Table 3. This is a direct check of 260 integer expressions. The table contains $(0,-1)$ and omits $(0,0)$, so Lemma 6.2 also excludes a witness for $k=0$.

Proof of Theorem 1.2. The certificates prove $M \notin \mathcal{V}$ for every $k \ge 0$. Since $M \equiv 1 \pmod 3$, it does not belong to $\mathcal{O}_3$: an element of $\mathcal{O}_3$ coprime to 3 would have to be congruent to 2 modulo 3. For $m \ge 3$,

$$2 \cdot 3^{m+2} < 23 \cdot 3^m - 20 < \tfrac{3}{2}\, 3^{m+3}.$$

These are the upper and lower endpoints of two consecutive intervals in the definition of $\mathcal{O}_\infty$, so $M \notin \mathcal{O}_\infty$. Hence $M \in \mathcal{F}$. The numbers $621 \cdot 3^{4k} - 20$ are distinct and unbounded, proving that $\mathcal{F}$ is infinite. □

9 An improved exponent without an epsilon loss

This section is new in version 2. The original power bound and the exceptional-family proof above are retained with their original numbering. Write $s = \log_3 2$, $\varphi = (1+\sqrt{5})/2$, and $g = \log_3 \varphi$.

Theorem 9.1. *There are absolute constants $c_* > 0$ and X_0 such that*

$$M_{\mathrm{C}}(X) \ge c_* X^{\beta_*} \quad (X \ge X_0), \qquad \beta_* = \frac{(1+s)^2}{7-5s} = 0.691726633148796\ldots.$$

The same lower bound holds for $M_{\mathrm{H}}(X)$. No subtraction of an arbitrary positive epsilon from β_* is required.

9.1 A boundary estimate for the mean Fourier discrepancy

Use $F_l, f_l, \mathcal{D}_l$ from Section 2. Their second and fourth moments are 2^{-l} and $(3/8)^l$. Let T_D be all nonzero reduced fractions on $\mathbb{T}$ with denominator at most D coprime to 3, and put $S_D(L) = \sum_{z \in T_D} f_L(z)$ and $h = \lceil 2\log_3 D \rceil$. Multiplication by 3 permutes T_D, whose separation is at least D^{-2}. The sampling lemma gives, uniformly for $0 \le m \le h$,

$$\sum_{T_D} f_m^2 \ll D^2 2^{-m}, \qquad \sum_{T_D} f_m^4 \ll D^2 (3/8)^m.$$

Indeed $3^h < 3D^2$, so the degrees in both applications are $O(D^2)$. Set

$$a_* = \frac{3s-1}{4}, \quad A_* = \frac{2(3-s)}{3s-1}, \quad \eta_0 = 3-4s, \quad r_* = \frac{3-s}{2} = a_* A_*.$$

Here $4 < A_* < 6$, $0 < \eta_0 < s$, and $r_* > 1$. For $2h \le L < 3h$ write $l = L - 2h$ and apply Holder's inequality with exponents $2, 4, 4$ to blocks of lengths h, h, l. This yields

$$S_D(L) \ll D^2 2^{-h/2} (3/8)^{(h+l)/4} \le C D^{(3+s)/2} 3^{-a_* L}. \tag{12}$$

The last inequality uses $s - 1 < 0$ and $h \ge 2\log_3 D$. For $L \ge 3h$, dropping extra factors instead gives

$$S_D(L) \le S_D(3h) \ll D^{\eta_0}. \tag{13}$$

For each fixed D, also $S_D(L) \to 0$: every nonzero fraction has a finite orbit under multiplication by 3, with all factors $|(1+\exp(2\pi i z))/2| < 1$.

Let $S_Y = \{N \le Y : N \in \mathcal{D},\ N \equiv 1 \pmod 3\}$ and $R_N(L) = \sum_{u=1}^{N-1} f_L(u/N)$.

Lemma 9.2. *There is an absolute C_* such that, for every fixed $H \geq 0$ and every integer choice $L = L(Y) \geq A_* \log_3 Y + H$,*

$$\limsup_{Y\to\infty} Y^{-s} \sum_{N\in S_Y} R_N(L) \leq C_* 3^{-a_* H}.$$

Proof. Fixing the high ternary digits shows that $T_Y(d) = \#\{N \in S_Y : d \mid N\} \leq 4(Y/d)^s$ for $1 \leq d \leq Y$: choose a low block with $3^j \leq d < 3^{j+1}$; its values lie in an interval of length less than d, hence contribute at most one multiple for each high block. Reducing u/N gives the exact identity

$$\sum_{N\in S_Y} R_N(L) = \sum_{\substack{2\leq d\leq Y\\(d,3)=1}} T_Y(d) \sum_{\substack{1\leq u<d\\(u,d)=1}} f_L(u/d).$$

In a fixed dyadic range $D/2 < d \leq D$, its contribution divided by Y^s is $O(D^{-s}S_D(L))$. One may take D to be powers of 2, with the final partial range bounded by $D < 2Y$. For ranges with $3h \leq L$, equation (13) supplies the summable majorant CD^{η_0-s}. Each fixed range tends to zero, so dominated convergence makes their total $o(1)$. For all remaining ranges, $2h \leq L$ once Y is sufficiently large, because $A_* > 4$. Equation (12) gives

$$D^{-s}S_D(L) \ll D^{r_*}3^{-a_*L} \leq 3^{-a_*H}(D/Y)^{r_*}.$$

The sum of $(D/Y)^{r_*}$ over these dyadic ranges is bounded absolutely. This proves the stated limsup. In particular the estimate asserts an arbitrarily small fixed proportion, not necessarily $o(Y^s)$ at fixed H. □

9.2 Affine tails and a fixed residue class

Lemma 9.3. *If $r \in 1 + 3\mathbb{Z}_3$, $r \neq 1$, $e = v_3(r-1)$, and $\xi \in \mathbb{Z}_3$, then*

$$\#\{z \in \mathcal{D}_m : [rz + \xi]_m \in \mathcal{D}_m\} \leq H_m(e) \leq \varphi^m(2/\varphi)^e,$$

where $[x]_m$ is the residue in $[0, 3^m)$ and $H_m(e)$ counts the independent sets of the graph on $0, \ldots, m-1$ with edges of length e.

Proof. Write $r = 1 + 3^e u$, with u a unit. In the addition $z + 3^e uz + \xi$, let B_j be the sum of the incoming carry, the digit of ξ, and the shifted digit of uz. On an admissible path the outgoing carry is $\lfloor (B_j + 1)/3 \rfloor$, independent of the admissible choice of z_j. Residues $B_j = 0, 1, 2 \pmod 3$ allow, respectively, two choices, only $z_j = 0$, or only $z_j = 1$. At position $i + e$, after fixing $z_{<i}$, the output digit is $z_{i+e} + u_0 z_i + Q_i(z_{<i}) \pmod 3$, with $u_0 \in \{1, 2\}$. Thus one of the two choices at i forces the digit at $i + e$, independently of intermediate admissible digits. Backward induction on the remaining positions bounds the tree by the independent-set recursion: at a free vertex one either removes that vertex, or removes it together with its future neighbor; an already forced vertex has only one branch. Discarding further restrictions only enlarges the bound. The graph splits into e chains, including empty chains. A chain of length l has $F_{l+2} \leq 2\varphi^{l-1}$ independent sets, also for $l = 0$. Multiplying proves the second inequality. □

Fix $c = 112 = (11011)_3$ and $d = 128$ throughout the rest of the proof. Define

$$C_d(c) = \#\{a, b, a', b' \in \mathcal{D}_d : c(a - a') + (b - b') \equiv 0 \pmod{3^d}\}.$$

For every right-hand side the corresponding count is at most $C_d(c)$, by Cauchy–Schwarz applied to the autocorrelation of $h(z) = \#\{(a, b) \in \mathcal{D}_d^2 : ca + b \equiv z \pmod{3^d}\}$. Also $C_d(c) \geq (16/3)^d$.

The following finite certificate specifies the constant used here without a numerical spectral-radius approximation. On states $-56,\dots,56$ put $T_{u,v}=w(3v-u)$, where

$$w(0)=4,\quad w(\pm1)=2,\quad w(\pm111)=1,\quad w(\pm112)=2,\quad w(\pm113)=1$$

and all other weights vanish. These are precisely the digit-difference multiplicities. The state interval is closed under every integral transition. Starting with $v_0(u)=\mathbf{1}_{u=0}$, iterate $v_{j+1}(v)=\sum_u v_j(u)T_{u,v}$. Then $C=C_d(c)=\sum_u v_{128}(u)$, with unrestricted final carry, equals

$$\begin{aligned}C=\,&284803932576710365592543571298693298054902154739229935778\\&5070566911039704321748559183306903592\end{aligned}$$

where the two lines are concatenated decimal digits. The exact integer checks supplied with the article give

$$C^{1161}3^{148608}\varphi^{491392}<2^{937216},\qquad 2^{20805}<3^{13127}. \tag{14}$$

These comparisons require integers only: for $m=491392$, write $\varphi^m=F_m\varphi+F_{m-1}$, multiply by $C^{1161}3^{148608}$ to obtain $A\varphi+B$, and check $D=2^{937217}-A-2B>0$ and $D^2>5A^2$. The executable certificate recomputes the matrix both forwards and backwards and performs these comparisons exactly. Set $\lambda=C^{1/d}$ and $\tau=\log_3\lambda$.

Lemma 9.4. *Let $Y=3^k$, $k>d$, and $G\subset\{N\in\mathcal{D}_k : N\equiv 1 \pmod{3^d}\}$. For $Y<X$ put $A_N(X)=\{1\le M\le X : M\equiv c \pmod{3^d},\ NM\in\mathcal{D}\}$. Then*

$$\sum_{N<N'\in G}|A_N(X)\cap A_{N'}(X)|\ll_d X^gY^{\tau-g}.$$

Proof. For four prefixes N',x,N,y of length $m\ge d$ with low blocks $1,c,1,c$ and $N'x\equiv Ny \pmod{3^m}$, adding four blocks of length d reduces the next congruence to

$$R+c(a-a')+(b-b')\equiv 0 \pmod{3^d},\qquad R=(N'x-Ny)/3^m.$$

The quadratic terms vanish since $m\ge d$. There are at most C extensions. If the first distinct denominator digit occurs at e, there are at most 4^{e-d} common prefixes and two choices at e. Applying full blocks and bounding a remaining partial block trivially by 16^d proves that the number of length-k quadruples is $O_d(4^e\lambda^{k-e})$.

Choose $n\ge k$ with $X<3^n\le 3X$. For each pair, the map $M\mapsto[NM]_n$ is injective, since N is a unit. Conditional on a prefix x_0 of length k, its remaining digits satisfy an affine condition with ratio $r=N'/N$ and shift $\xi=(rx_0-[rx_0]_k)/3^k$. Lemma 9.3 bounds these tails. Summing over e therefore gives

$$O_d\left(\varphi^{n-k}\lambda^k\sum_{e\ge d}\left(\frac{8}{\varphi\lambda}\right)^e\right).$$

The ratio is at most $3/(2\varphi)<1$, since $\lambda\ge 16/3$. This proves the assertion, including arbitrarily close distinct pairs. □

9.3 The first moment and completion of the proof

Put

$$\alpha_*=\frac{1}{A_*-1}=\frac{3s-1}{7-5s},\qquad Y=3^k=3^{\lfloor\log_3(c_0X^{\alpha_*})\rfloor},$$

where $c_0>0$ will be fixed before $X\to\infty$. The shell $\mathcal{S}_{k,d}=\{N\in\mathcal{D}_k : N\equiv 1 \pmod{3^d},\ N\ge 3^{k-1}\}$ has exactly $2^{-d-1}Y^s$ members, all in $[Y/3,Y/2)$. Choose a fixed H with $C_*3^{-a_*H}<$

2^{-d-5}, and then $c_0 \le 3^{-(H+d+4)/(A_*-1)}$. For $L_* = \lfloor \log_3(XY/3) \rfloor - d$ one has $L_* - A_* \log_3 Y \ge H+2$. Lemma 9.2 and Markov's inequality show that at least half the shell, denoted G, has $R_N(L_*) \le 1/2$ for all sufficiently large X.

For $L_N = \lfloor \log_3(NX) \rfloor$, count multiples of N of the form $z = c + 3^d w$, $w \in \mathcal{D}_{L_N-d}$. Orthogonality, permutation of residues by 3^d, and monotonicity of f_l give at least

$$\frac{2^{L_N-d}}{N}(1 - R_N(L_N - d)) \ge 2^{-d-2} X^s N^{s-1}$$

such multiples. They are positive and less than NX, and z/N lies in the required residue class. Consequently

$$\sum_{N \in G} |A_N(X)| \gg_d X^s Y^{2s-1}. \tag{15}$$

The exact inequalities (14) imply $\alpha_* < \alpha_0 = 1161/5000$ and $g - s + \alpha_0(1 + \tau - g - 2s) < 0$. The coefficient $1 + \tau - g - 2s$ is positive, since $\tau \ge 4s - 1$ and $2s > g$. Hence

$$g - s + \alpha_*(1 + \tau - g - 2s) < 0.$$

Lemma 9.4 is thus $o(1)$ times (15). Bonferroni's inequality gives

$$M_{\mathrm{C}}(X) \ge \left| \bigcup_{N \in G} A_N(X) \right| \gg X^s Y^{2s-1} \gg X^{s+(2s-1)\alpha_*} = X^{\beta_*}.$$

The positive-witness inclusion proved above transfers the bound to M_{H}. This completes the proof of Theorem 9.1. The constants can be very small and the threshold very large; no practical search threshold or sharp asymptotic is asserted.

10 Positive dimension for the sporadic family

The absence of a positive finite integer witness does not imply that the corresponding 3-adic intersection has dimension zero. We now determine this distinction for every multiplier in Theorem 1.2. Put

$$M_k = 621 \cdot 3^{4k} - 20, \qquad (m_k, L_k) = \begin{cases} (3, 12), & k = 0, \\ (4k+3, 12k+6), & k \ge 1. \end{cases}$$

We use $\mathcal{H} = \{M \in \mathbb{N} : \dim_H C(1, M) > 0\}$ for the multiplier set counted by $M_{\mathrm{H}}(X)$; $\mathcal{V}$ remains the set of positive finite-integer quotients counted by $M_{\mathrm{C}}(X)$.

Theorem 10.1 (Dimension of an explicit subsystem)**.** *For every $k \ge 0$, let $d_k > 0$ be the unique solution of*

$$3^{-m_k d_k} + 3^{-L_k d_k} = 1. \tag{16}$$

There is a compact subset of $C(1, M_k)$ of Hausdorff dimension exactly d_k. Consequently

$$\dim_H C(1, M_k) \ge d_k > 0, \qquad M_k \in \mathcal{H} \setminus \mathcal{V}.$$

If $\vartheta > 1$ is the positive root of $\vartheta^4 - \vartheta^3 - 1 = 0$, then

$$d_k \ge \frac{\log_3 \vartheta}{m_k}, \tag{17}$$

with equality for $k = 0$ and strict inequality for $k \ge 1$.

The exact value in (16) concerns the constructed subsystem, not all of $C(1, M_k)$. The conclusion that these multipliers are not finite-integer quotients uses Theorem 1.2; the positive-dimension construction below is independent of its proof. The distinction between positive dimension and finite witnesses is already present in Lagarias's work [1], including the example $M = 52$. Our assertion is the explicit construction and dimension bound for this particular sporadic family proposed in [4, §1.4].

10.1 Words and carry paths

In this section a word $w = w_0 \cdots w_{\ell-1}$ is written *least significant digit first*, with

$$|w| = \ell, \qquad [w] = \sum_{j=0}^{\ell-1} w_j 3^j.$$

Juxtaposition denotes concatenation, so $[uv] = [u] + 3^{|u|}[v]$. All displayed input and output words have digits in $\{0, 1\}$. A carry transition for multiplication by $M \in \mathbb{N}$ is

$$Ma_j + c_j = b_j + 3c_{j+1}. \tag{18}$$

Lemma 10.2 (An integer identity certifies a path)**.** *Let $c, d \geq 0$ be integers, and let w, v be binary words of the same length ℓ. If*

$$M[w] + c = [v] + 3^\ell d, \tag{19}$$

then digitwise multiplication gives an admissible path from carry c to carry d, with input w and output v.

Proof. Reduction modulo 3 identifies the first output digit with v_0. Since $Mw_0 + c \geq 0$, the quotient after removing this residue is a nonnegative integer. Subtract the first-digit equality from (19) and divide by 3. Repetition proves the claim at every digit, including the prescribed final carry. □

Suppose a fixed admissible prefix from carry 0 reaches c, and two words A, B return c to itself. Any infinite concatenation of these two words, appended to the prefix, is an admissible input $x \in \Sigma$. Its output digits also lie in $\{0, 1\}$, and they represent Mx: telescoping (18) through the first n digits leaves the error $3^n c_n$, which tends to zero in $\mathbb{Z}_3$ because c_n is an integer. Thus the whole construction lies in $C(1, M)$. It is essential here that the return carry is reachable from 0.

10.2 The initial multiplier

For $k = 0$ we have $M_0 = 601$. Take $c = 69$ and the following paths:

	input	output	carries
prefix	1110010010	1011001101	$0 \longrightarrow 69$
A	010	001	$69 \longrightarrow 69$
B	101110010010	110101101101	$69 \longrightarrow 69$

The respective input values are $6817, 3, 61354$, and the output values are $22636, 9, 204394$. The three exact identities are

$$601 \cdot 6817 = 22636 + 69 \cdot 3^{10},$$
$$601 \cdot 3 + 69 = 9 + 69 \cdot 3^3,$$
$$601 \cdot 61354 + 69 = 204394 + 69 \cdot 3^{12}.$$

Lemma 10.2 therefore supplies a reachable carry with two return words of lengths 3 and 12, starting with different digits.

10.3 A reachable carry for every positive parameter

Fix $k \geq 1$ and write

$$t = 81^k, \quad m = 4k+3, \quad L = 12k+6, \quad M = 621t - 20 = 23 \cdot 3^m - 20, \quad c = 69t.$$

The prefix

$$P = (1100)^k \, 111 \, 0^{4k} \, 1 \, 0^{4k+2} \tag{20}$$

has length L. Its value is

$$p = [P] = \frac{t-1}{20} + 13t + 27t^2 = \frac{540t^2 + 261t - 1}{20}.$$

Using $3^L = 729t^3$, direct multiplication gives

$$q := Mp - 23 \cdot 3^L = \frac{151281t^2 - 5841t + 20}{20}. \tag{21}$$

We next verify that q is the value of a binary ternary word of length L, allowing zeros at its high end. For $k = 1$, its base-81 digits in least-significant-first order are

$$(1, 36, 27, 12, 1).$$

For $k \geq 2$ they are

$$(1, \underbrace{0, \ldots, 0}_{k-1}, 36, 0, \underbrace{4, \ldots, 4}_{k-2}, 31, 12, 1).$$

Indeed, the latter digit list is the identity

$$\begin{aligned} q = 1 + 36 \cdot 81^k + 4 \sum_{j=k+2}^{2k-1} 81^j \\ + 31 \cdot 81^{2k} + 12 \cdot 81^{2k+1} + 81^{2k+2}, \end{aligned}$$

which follows from the geometric sum formula and (21); an empty sum is zero. Every listed base-81 digit belongs to $\{0, 1, 4, 12, 27, 31, 36\}$ and hence has four ternary digits in $\{0, 1\}$ after padding. The highest nonzero base-81 digit is 1 in position $2k+2$, so the ternary length is at most $8k + 9 \leq L$. Thus P reaches carry 23.

The input 10 with output 01 then reaches c, since

$$M + 23 = 3 + 9c. \tag{22}$$

Consequently the binary prefix $P10$ reaches $c = 69t$ from the initial carry 0.

10.4 Two return cycles

The short cycle is

$$A = 0^{m-2}10, \qquad |A| = m, \qquad [A] = 3^{m-2} = 3t.$$

Its output is $0^{m-1}1$, because

$$M[A] + c = 3^{m-1} + 3^m c. \tag{23}$$

For the long cycle, put

$$B = (1100)^{k-1} \, 11010110 \, (0110)^{k-1} \, 01 \, 0^{4k+2} \, 10. \tag{24}$$

The lengths of its six successive pieces sum to L. To compute its value, write $r = 81^{k-1} = t/81$ and use concatenation:

$$\begin{aligned} b = [B] &= \frac{r-1}{20} + 1003r + \frac{19683}{20}r(r-1) + 3t^2 + 81t^3 \\ &= 81t^3 + \frac{63}{20}t^2 + \frac{7}{30}t - \frac{1}{20}. \end{aligned}$$

Here $[11010110] = 1003$ and $[0110] = 12$; the two geometric sums account for the repeated four-digit pieces. The required output value is

$$z := Mb + c - 3^L c = \frac{20169t^3 + 4914t^2 + 1997t + 60}{60}. \tag{25}$$

For every $k \geq 1$, the geometric sum formula gives the exact base-81 expression

$$\begin{aligned} z = 1 + 27 \cdot 81^{k-1} + 37 \cdot 81^k + 4 \sum_{j=k+1}^{2k-1} 81^j \\ + 13 \cdot 81^{2k} + 13 \cdot 81^{2k+1} + 12 \sum_{j=2k+2}^{3k} 81^j + 4 \cdot 81^{3k+1}. \end{aligned}$$

For $k \geq 2$ the nonoverlapping terms give the digit list

$$(1, \underbrace{0, \ldots, 0}_{k-2}, 27, 37, \underbrace{4, \ldots, 4}_{k-1}, 13, 13, \underbrace{12, \ldots, 12}_{k-1}, 4).$$

For $k = 1$ only the first two terms overlap, combining to 28; the list is $(28, 37, 13, 13, 4)$. All the digits in either list belong to $\{0, 1, 4, 12, 13, 27, 28, 37\}$, each of which has only $0, 1$ as ternary digits. The highest digit $4 = (11)_3$ is in base-81 position $3k+1$ and occupies ternary positions $L-2, L-1$. Therefore $z < 3^L$ and z is a valid length-L output word.

Lemma 10.2, applied to (23) and (25), proves that both A and B return to c. Their first digits are respectively 0 and 1. These identities and digit patterns apply to all $k \geq 1$; no extrapolation from finitely many carry graphs is used.

10.5 Dimension of the return-word construction

We give the dimension argument explicitly, including the differing word lengths. For either construction above, let $K \subset \mathbb{Z}_3$ be the set of all infinite concatenations of A and B without the fixed entry prefix. This is compact, and

$$K = ([A] + 3^m K) \ \cup \ ([B] + 3^L K).$$

The two pieces have different first ternary digits. In particular, the two-word code is prefix-free, its infinite addresses are unique, and these similarities have disjoint images on K. Let $d > 0$ solve $3^{-md} + 3^{-Ld} = 1$. Existence and uniqueness follow from strict monotonicity: the left side decreases from 2 to 0 as d increases from 0 to infinity.

Give the independent choices A, B probabilities 3^{-md} and 3^{-Ld}. The induced measure μ on K assigns mass 3^{-ds} to an address cylinder whose completed blocks have total ternary length s. Consider a 3-adic ball of radius 3^{-n} that meets K, so its first n ternary digits are fixed. The distinct first digits of A and B allow one to parse from these digits every completed block before depth n. That completed-block prefix is shared by all addresses in the ball. Its length s satisfies $s \geq n - L$, since each block has length at most L. Hence

$$\mu(B) \leq 3^{-ds} \leq 3^{dL} 3^{-dn}. \tag{26}$$

The mass distribution principle yields $\dim_H K \geq d$.

For the reverse inequality, stop each address at the first completed block of total length at least n. The stopped lengths lie in $[n, n+L-1]$. These finitely many address cylinders partition the address space, so the sum of their masses is 1. Their image diameters are at most 3^{-s}, and thus the sum of the dth powers of those diameters is at most 1. They cover K at arbitrarily small scales. It follows that $\mathcal{H}^d(K) \leq 1$ and $\dim_H K \leq d$.

Appending the reachable entry prefix maps K by a fixed 3-adic similarity into $C(1, M_k)$, and therefore preserves its dimension. We have proved the subsystem assertion in Theorem 10.1; Theorem 1.2 supplies $M_k \notin \mathcal{V}$.

Finally $L_k \leq 4m_k$, with equality only for $k = 0$. Since $\vartheta^{-1} + \vartheta^{-4} = 1$, at $d = (\log_3 \vartheta)/m_k$ the left side of (16) is at least 1, and is strictly larger for $k \geq 1$. Monotonicity proves (17) and its asserted strictness.

Remark 10.3 (Scope). This result describes an explicit positive-dimensional subset for every member of the sporadic nonquotient family. Its lower bound depends on k; no uniform positive lower bound or exact dimension of the full intersection is asserted. It neither determines the sharp asymptotics of $M_{\mathrm{C}}(X)$ nor excludes the point 1 from the exceptional set for powers of two.

11 A universal polynomial quotient at 289

The distinction between finite and infinite digit expansions has a second algebraic aspect. For nonzero polynomials $P, Q \in \mathbb{Z}[t]$ with coefficients in $\{0, 1\}$, a representation $M = P(3)/Q(3)$ is called *universal* if Q divides P, and *local* otherwise. This is the terminology of [5, Definition 1.1], after removing any common initial power of t. Indeed, for $M \equiv 1 \pmod 3$ the least nonzero digits of $P(3)$ and $Q(3)$ occur at the same position; removing these equal powers of t gives constant terms 1, without changing polynomial divisibility. Since Q is monic, divisibility in $\mathbb{Q}[t]$ is equivalent to divisibility in $\mathbb{Z}[t]$. The distinction concerns each representation, rather than just its integer value.

The catalogue in [5, Section 4, pp. 24–25], specifically arXiv version 1, leaves open the existence of a local indecomposable representation of 289. We answer this question by proving a stronger statement about all finite representations, without an indecomposability restriction.

Theorem 11.1 (A common polynomial quotient for every representation)**.** *Put*

$$h(t) = 1 - t^2 - t^3 + t^4 + t^5.$$

For all nonzero polynomials $P, Q \in \mathbb{Z}[t]$ with coefficients in $\{0, 1\}$,

$$P(3) = 289Q(3) \quad \Longrightarrow \quad P(t) = h(t)Q(t). \tag{27}$$

In particular, every finite representation of 289 is universal, and there is no local representation of 289.

11.1 Returning carry paths and polynomial potentials

We give the finite certificate and its all-length interpretation explicitly. Let $M > 1$ and $M \equiv 1 \pmod 3$. An input digit a and an output digit b in $\{0, 1\}$ give the carry transition

$$3c' = c + Ma - b, \qquad c_0 = 0. \tag{28}$$

Along a path, induction gives

$$3^j c_j = M \sum_{i<j} a_i 3^i - \sum_{i<j} b_i 3^i. \tag{29}$$

All reachable carries are nonnegative: the only negative numerator possible from $c \geq 0$ is $c = a = 0, b = 1$, and it is not divisible by 3. The invariant also gives $c_j < M/2$, since $\sum_{i<j} a_i 3^i \leq (3^j - 1)/2$. For each nonnegative carry all integral transitions are exactly

$c \pmod 3$	(a, b, c')
0	$(0, 0, c/3), \quad (1, 1, (c + M - 1)/3),$
1	$(0, 1, (c - 1)/3),$
2	$(1, 0, (c + M)/3).$

(30)

These are obtained by checking the four digit pairs modulo 3.

Every finite pair $P(3) = MQ(3)$, padded with high zero digits to a common length n, determines a path from 0 to 0: its carries are integers because each prefix difference is divisible by 3^j. Conversely, a path from 0 to 0 gives this integer equality by (29). At least one input digit is 1 exactly when the witness is positive.

Lemma 11.2 (Polynomial potential certificate)**.** *Suppose S contains every state on every path from 0 back to 0. Let $h \in \mathbb{Z}[t]$ satisfy $h(3) = M$. Suppose polynomials $H_c \in \mathbb{Z}[t]$, for $c \in S$, satisfy $H_0 = 0$ and*

$$tH_{c'} = H_c + ah - b \tag{31}$$

on every transition $c \xrightarrow{a,b} c'$ inside S. Then every finite witness satisfies $P = hQ$.

Proof. Multiply (31) on the jth edge by t^j and sum along a returning path of length n. The potentials telescope:

$$t^n H_{c_n} - H_{c_0} = h \sum_{j<n} a_j t^j - \sum_{j<n} b_j t^j.$$

The left side is zero. This is an identity in $\mathbb{Z}[t]$ for each finite n, with no bound on the witness length. □

11.2 The complete certificate for 289

For $M = 289$, define the disjoint sets

$$\begin{aligned} S = \{&0, 1, 4, 11, 12, 32, 33, 37, 44, 48, 96, 100, 107, 111, 112, \\ &132, 133, 140, 143, 144\}, \\ D = \{&5, 14, 15, 16, 43, 46, 98, 101, 128, 129, 130, 139\}. \end{aligned}$$

Table 4 lists all transitions out of $S \cup D$, using (30), and gives the polynomials H_c on S. A dash marks a state in D. Every listed destination is in $S \cup D$, so this set contains all states reachable from 0. Every transition out of D stays in D, and $0 \notin D$. Consequently a returning path cannot enter D; it stays in S.

For each of the 26 edges with both endpoints in S, the displayed polynomials satisfy (31). This is checked by subtracting the two sides coefficient by coefficient, with $h = 1 - t^2 - t^3 + t^4 + t^5$. For example, the edges from 0, 4 and 144 give

$$tH_{96} = h - 1, \qquad tH_1 = H_4 - 1, \qquad tH_{144} = H_{144} + h - 1.$$

The table is therefore a finite, explicit certificate for every hypothesis of Lemma 11.2. Merely checking $H_c(3) = c$ would not suffice; the full polynomial identities are the input to the lemma.

Proof of Theorem 11.1. We have $h(3) = 289$ and $H_0 = 0$. The preceding closure checks show that all returning paths stay in S, and the table verifies the potential identity on every edge they can use. Apply Lemma 11.2. □

Carry c	Potential $H_c(t)$	All transitions $(a,b):c'$
0	0	$(0,0):0$; $(1,1):96$
1	1	$(0,1):0$
4	$1+t$	$(0,1):1$
5	–	$(1,0):98$
11	$-1+t+t^2$	$(1,0):100$
12	$t+t^2$	$(0,0):4$; $(1,1):100$
14	–	$(1,0):101$
15	–	$(0,0):5$; $(1,1):101$
16	–	$(0,1):5$
32	$-1-t+t^2+t^3$	$(1,0):107$
33	$-t+t^2+t^3$	$(0,0):11$; $(1,1):107$
37	$1+t^2+t^3$	$(0,1):12$
43	–	$(0,1):14$
44	$-1+2t^2+t^3$	$(1,0):111$
46	–	$(0,1):15$
48	$t+2t^2+t^3$	$(0,0):16$; $(1,1):112$
96	$-t-t^2+t^3+t^4$	$(0,0):32$; $(1,1):128$
98	–	$(1,0):129$
100	$1-t^2+t^3+t^4$	$(0,1):33$
101	–	$(1,0):130$
107	$-1+t^3+t^4$	$(1,0):132$
111	$t+t^3+t^4$	$(0,0):37$; $(1,1):133$
112	$1+t+t^3+t^4$	$(0,1):37$
128	–	$(1,0):139$
129	–	$(0,0):43$; $(1,1):139$
130	–	$(0,1):43$
132	$-t+2t^3+t^4$	$(0,0):44$; $(1,1):140$
133	$1-t+2t^3+t^4$	$(0,1):44$
139	–	$(0,1):46$
140	$-1-t+t^2+2t^3+t^4$	$(1,0):143$
143	$-1+t^2+2t^3+t^4$	$(1,0):144$
144	$t^2+2t^3+t^4$	$(0,0):48$; $(1,1):144$

Table 4: A closed carry certificate for $M = 289$. The 32 rows list all 42 edges out of the displayed states. Potentials are assigned exactly on S; the other 12 states form the closed set D.

The theorem is not vacuous. Direct multiplication gives

$$(1 - t^2 - t^3 + t^4 + t^5)(1 + t^2 + t^3 + t^5) = 1 + t^7 + t^9 + t^{10}, \tag{32}$$

which at $t = 3$ is $289 \cdot 280 = 80920$. The accompanying verifier independently constructs all allowable transitions in the a priori range $0 \le c \le 144$, checks closure and all polynomial coefficients, and checks (32). Its tests of additional long returning paths are regression tests; the all-length conclusion follows from the displayed certificate and telescoping argument.

11.3 An explicit family of local finite representations

The same polynomial viewpoint produces a family for which a specified representation is local. To distinguish it from the sporadic family M_k in Theorem 1.2, write

$$N_k = 4 \cdot 3^k - 2 \qquad (k \ge 1).$$

Theorem 11.3 (Explicit finite witnesses)**.** *For every $k \ge 1$, $N_k \in \mathcal{V}$. There are $0,1$ polynomials P_k, Q_k with*

$$P_k(3) = N_k Q_k(3), \qquad \deg Q_k = k^2 - 1, \qquad \deg P_k = k(k+1).$$

For $k \ge 2$, the representation specified below is local.

Proof. Set

$$Q_k(t) = \sum_{j=0}^{k-1} t^{(k+1)j} \sum_{i=0}^{k-j-1} t^i. \tag{33}$$

The exponent intervals in the outer sum are disjoint, and the final one ends at $(k+1)(k-1) = k^2 - 1$. Thus Q_k is a $0,1$ polynomial of the asserted degree, with constant term 1. For $k = 1$, take $P_1 = 1 + t^2$ and $Q_1 = 1$. For $k \ge 2$, put $n = k(k+1)$ and set

$$P_k(t) = 1 + \sum_{i=2}^{k} t^{ki} + \sum_{j=1}^{k-2} t^{k+(k+1)j} + t^{n-1} + t^n. \tag{34}$$

The exponents in the second sum have distinct nonzero residues $1, \dots, k-2$ modulo k; the exponents in the first sum are multiples of k. The final two exponents exceed all preceding ones. Hence all displayed exponents are distinct, so P_k is also a $0,1$ polynomial, of degree n.

Write $S_k(t) = \sum_{i=0}^{k-1} t^i$. Shifting the blocks in (33) and subtracting successive blocks gives

$$(t^{k+1} - 1)Q_k = \sum_{i=2}^{k+1} t^{ki} - S_k. \tag{35}$$

A second identity is

$$(t^k - 1)Q_k = \sum_{j=0}^{k-1} t^{k+(k+1)j} - S_k. \tag{36}$$

For completeness, multiplying (33) by $t - 1$ gives

$$(t-1)Q_k = \sum_{i=1}^{k} t^{ki} - \sum_{j=0}^{k-1} t^{(k+1)j}.$$

Multiplication by $t^k - 1$, followed by

$$(t^k - 1)\sum_{i=1}^{k} t^{ki} = t^{k(k+1)} - t^k, \qquad (t^{k+1} - 1)\sum_{j=0}^{k-1} t^{(k+1)j} = t^{k(k+1)} - 1,$$

proves (36) after cancellation of the nonzero polynomial $t-1$. Adding (35) and (36), and separating their extreme terms, now gives

$$\begin{aligned} P_k &= (t^{k+1}+t^k-2)Q_k + R_k, \\ R_k &= 3 + 2\sum_{i=1}^{k-1} t^i - t^k. \end{aligned} \tag{37}$$

At $t=3$, the remainder is zero, so $P_k(3) = (4\cdot 3^k - 2)Q_k(3)$. But R_k is a nonzero polynomial with $\deg R_k = k < k^2 - 1 = \deg Q_k$ for every $k \geq 2$. Thus (37) is polynomial division with a nonzero remainder. The displayed representation is local. □

Remark 11.4 (Scope of the two conclusions). Theorem 11.1 determines the type of *every* representation of one integer. Theorem 11.3 constructs *one* representation for each member of an infinite family; it does not exclude other universal representations or claim minimal witness length. The output length is $k(k+1)+1$, hence $O((\log N_k)^2)$ on this family. Since the family has only $O(\log X)$ members up to X, this construction does not improve the counting exponent of Theorem 9.1. Neither conclusion proves the finiteness of good powers of two.

12 Verification and scope

The two new sections received independent internal mathematical reviews of their final English sources. The family program checks seven exact symbolic identities and digitwise examples; the all-parameter dimension argument is supplied in the text. The numerator program independently reconstructs the complete reachable carry graph, verifies the nonreturn component and every coefficient of all internal polynomial-potential identities. It also tests the explicit local family. The finite graph certificate, together with telescoping, covers returning paths of every length; sampling long paths alone would not establish this conclusion. The older proof sections and their review records retain their original scope.

For the version-2 bound, `code/endpoint/run_checks.py` independently recomputes the exact 113-state energy forwards and backwards, checks the integer sign certificates, and tests block lifting, affine tails, and endpoint algebra. These are finite inputs and consistency checks; the limiting estimates are proved in Section 9. The underlying archived arguments had internal reviews; the new English section received a focused integration review. The retained version-1 proof bodies are not represented as freshly re-reviewed.

All estimates for the power lower bound, including the required uniformity and limiting arguments, are proved in the text. The finite-word estimate counts every locally admissible word; it does not assume an infinite continuation. The accompanying elementary program `verify_power_gain_lemmas.py` checks exact finite digital energies, the finite-word bound for unit ratios modulo 3^n through $n=9$, and the stated divisor-cylinder and valuation-bucket bounds at finite scales. These checks supplement the proofs. They cannot establish their asymptotic conclusions by themselves.

The nonrepresentation proof consists of the delayed-digit argument and the explicit predecessor certificates. Its parameter-dependent part is a symbolic calculation, rather than an extrapolation from finitely many tested values of k. The program `verify_parametric_certificate.py` checks every carry image in Table 2. Splitting the combined rows into interior and boundary variants gives 18 symbolic rows and 384 choices of carry and binary digits. The register identities are checked as affine polynomials in formal variables $X = 16^k$ and $Y = T_j$, with exact rational coefficients. The most significant bit inequalities are proved in Proposition 7.1. The program also checks all 260 predecessor choices in the separate $k=0$

certificate and makes redundant integer instantiations to detect indexing or transcription errors. Both programs use only the Python standard library and exact integer or rational arithmetic.

Theorems 9.1 and 1.2 give a power lower bound and an explicit infinite sporadic family, respectively. They do not give a full asymptotic, a density theorem, or a classification of all nonrepresentable multipliers. The known positive-witness construction is sufficient, not necessary, for positive Hausdorff dimension.

ChatGPT was used in developing and editing the arguments, implementing checks, and conducting internal AI-assisted reviews from several mathematical perspectives. Those reviews are not external peer review or proof-assistant verification. Authorship is attributed solely to Timofei Iuzhakov.

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